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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Fundamentals of Object Oriented Programming (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

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Week 4 ()

Thank you for taking the Week 4: Assignment 4.

Week 4: Assignment 4

Your last recorded submission was on 2026-02-17, 21:04 IST Due date: 2026-02-18, 23:59 IST.

1) Which of the following best describes polymorphism in object-oriented programming? ***1 point***

- A class having multiple constructors.
- The ability of different objects to respond to the same function call in different ways.
- A function having the same name as its class.
- None of these.

2) Which of the following is an example of static polymorphism? ***1 point***

- Method overloading
- Virtual functions
- Abstract classes
- Method overriding

3) Consider the following code: ***1 point***

```
1 1 1class Complex {2    int real, imag;3public:4    Complex(int r, int i) :
```

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Polymorphism (unit?unit=42&lesson=51)
Overloading: Operator and Constructor (unit?unit=42&lesson=52)
'this' keyword in C++ (unit?unit=42&lesson=53)
Method Overloading (unit?unit=42&lesson=54)
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What is the output of this program?

- (selected) 6 + 8i
- 6 + 15i
- 8 + 8i
- Compilation error

4) In a C++ program to overload the * operator for a class Matrix, where: ***1 point***

- The class stores a 2D matrix as a private member.
- The * operator multiplies two matrices.
- The result of multiplication is displayed in the console.

Which of the following correctly implements the operator overloading?

- The operator is defined inside the class.
- The operator is defined as a friend function.
- (selected) Both A and B are valid.
- Operator overloading is not possible for matrix multiplication.

5) Consider the following Java code: ***1 point***

```
1 1•x 1class Calculator {2    int add(int a, int b) {3        return a + b;4
```

What is the output of this program?

- (selected) 5 6.0
- 5.0 6.0
- Compilation error
- 5 5

6) Which of the following demonstrates dynamic polymorphism? ***1 point***

```
1 •x 1class Base {2public:3    virtual void display() { std::cout << "Base clas
```

What is the output of this program?

- Base class
- (selected) Derived class
- Compilation error
- Undefined behavior

7) Which of the following is true about virtual functions in C++? ***1 point***

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- (●) They allow runtime polymorphism.
- ( ) They must be redefined in the derived class.
- ( ) They can be called on an object of the base class.
- ( ) They cannot be used with pointers.

8) Consider the following Java code: ***1 point***

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }
}

public class Main {
    public static void main(String[] args) {
        Animal animal = new Dog();
        animal.sound();
    }
}
```

What is the output of this program?

- ( ) Animal makes a sound
- (●) Dog barks
- ( ) Compilation error
- ( ) Undefined behavior

9) Which of the following is a limitation of static polymorphism? ***1 point***

- ( ) It requires pointers.
- (●) It is resolved at compile time and cannot adapt to runtime behavior.
- ( ) It can only be implemented in C++.
- ( ) It cannot be overloaded.

10) In a C++ program to demonstrate both static and dynamic polymorphism using the following: ***1 point***

- Method overloading for static polymorphism.
- Virtual functions for dynamic polymorphism.

Which of the following correctly calls both overloaded and overridden methods?

- (●) Overloaded methods are called directly, and overridden methods are called using a base class pointer.
- ( ) Overloaded methods are called using a base class pointer, and overridden methods are called directly.

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- ◯ Both methods are called directly.
- ◯ Both methods require pointers.

You may submit any number of times before the due date. The final submission will be considered for grading.

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